

GENDER DIFFERENCE IN MEMORY FUNCTIONS FOLLOWING LONG TERM TRYPTOPHAN SUPPLEMENTATION IN RATS

**SAIMA KHALIQ, SAIDA HAIDER, ANA MUKHTAR
AND DARAKHSHAN J. HALEEM**

*Department of Biochemistry,
Neurochemistry and Biochemical Neuropharmacology Research Unit
University of Karachi, Karachi-75270, Pakistan*

ABSTRACT:

Evidence shows that tryptophan (TRP) and serotonin (5HT) have a role in memory function. It has been shown in various studies that increase in serotonergic neurotransmission is associated with increased memory consolidation whereas low brain 5HT impairs memory performance. In view of a possible role of TRP and 5HT in memory, the present study was designed to investigate the effect of TRP supplementation on male and female rats and to investigate sex related differences in memory processes. Tryptophan at a dose of 100mg/kg was used. Short term memory (STM) and long term memory (LTM) were evaluated using the Morris water maze. TRP treatment for 6 weeks significantly enhanced STM of female rats but it did not have any effect on the STM of male rats. Female rats when compared with male rats exhibited better performance in the Morris water maze and enhanced memory was observed both before and after TRP treatment, however after TRP treatment a greater improvement was observed. Both male and female rats exhibited improvement in LTM following TRP treatment. The present results emphasize on the involvement of sex difference and 5HT in learning and memory processes. A greater increase in 5HT metabolism and turnover seen in female rats correlates with the enhanced memory function observed in females than male rats.

INTRODUCTION

It is well known that the synthesis of 5HT in the adult brain is dependent upon the availability of its precursor amino acid TRP to the serotonergic neurons (Leathwood *et al.*, 1987; Fernstrom *et al.*, 1994). This occurs because the rate limiting enzyme in the biosynthetic pathway of 5HT, TRP hydroxylase located in the serotonergic neuron (Aghajanian and Asher, 1971) is unsaturated with its substrate under normal physiological conditions. TRP loading increases plasma TRP/ LNAAs (large neutral amino acids) ratio and it increases brain 5HT and 5HIAA levels and therefore increases brain serotonin activity (Markus *et al.*, 2002). It has been shown in various studies that increase in serotonergic neurotransmission is associated with increased

memory consolidation (Harmer *et al.*, 2002) whereas low brain 5HT impairs memory performance in humans (Hughes *et al.*, 2002; Porter *et al.*, 2003) and animals (Lieben *et al.*, 2004).

Sex difference in central serotonergic system has been reported in many neurochemical and behavioral studies. 5HT synthesis and turnover in whole brain and many brain regions also showed sex difference of about 20-25%. A number of neurochemical investigations have consistently shown that 5HT metabolism and turnover was greater in female than in male rats (Haleem, 1992; Haleem *et al.*, 1990). However increased 5HT metabolism did not necessarily imply greater 5HT release at the functional postsynaptic site. If the greater 5HT synthesis is associated with

greater release of 5HT to receptors, this is likely to lead to sex difference in various brain functions.

The investigation of sex differences in human spatial cognition has a long history and includes many components of spatial ability. Males perform better than females in mental rotation tasks (Tapley and Briden, 1977). Females have better memory for object location in static (Silverman and Eals, 1992) and in dynamic environments (Tottenham *et al.*, 2003). Several animal studies have also shown a sex difference in memory functions favoring males (Beatty, 1979; Gresack and Frick, 2003) others favored females (Bimonte *et al.*, 2000) or no sex difference was reported (Bucci *et al.*, 1995).

Previously we have also shown that administration of TRP increased brain 5HT metabolism (Khaliq *et al.*, 2006). Based on the above consideration, the present study was therefore designed to investigate the effects of long term TRP administration on the memory function in male and female rats, in order to investigate sex related differences in memory processes.

MATERIAL AND METHODS

Twenty-four locally bred albino Wister rats (120-130 g) purchased from Agha Khan University hospital were used in the study. All animals were housed individually under a 12 hour light-dark cycle (light on at 6:00 h) and controlled room temperature ($22 \pm 2^\circ\text{C}$) with free access to cubes of standard rodent diet and tap water for at least 3-4 days before experimentation so that rats could adapt themselves to the new environment. All experiments were conducted according to a protocol approved by Local Animal Care Committee.

Drug Administration:

Tryptophan at a dose of 100mg/kg b.w was used. The experimental protocol was designed to administer tryptophan orally with

the help of feeding tube for 6 weeks. Male and female rats were divided into control and test group. Test groups received 100mg/kg body weight TRP daily for 6 weeks. Weighed amount of food was placed in the hopper of all the cages. Body weights and food intakes were monitored weekly. Behavioral activities of rats were also monitored. Rats were decapitated after 6 weeks between 10:00 and 11:00 h to collect plasma and brain samples. The experiment was performed in a balanced design in such a way that control and drug treated rats were killed alternately to avoid the order effect. After decapitation blood was collected in heparinized tubes and centrifuged to get plasma. These plasma samples were then stored below -70°C for estimation of TRP. Brain samples were excised very quickly from the cranial cavity within 30 seconds of the decapitation. Fresh brains were dipped in chilled saline (0.9% w/v) and stored at low temperature (-70°C) until analysis of 5HT, 5HIAA and TRP by HPLC-EC as described earlier (Haleem and Perveen, 1994).

BEHAVIORAL TEST

Morris Water Maze:

Water maze is the classic experimental procedure first described by Morris (Morris, 1981) (Fig.1). Spatial memory was determined by noting the latency time (time in seconds taken by the rats to reach the platform). Maze consisted of rectangular pool (60×30 cm) filled with water (27°C), made opaque by addition of milk, to the depth of 12 cm. Water temperature had to be accurate to $\pm 1^\circ\text{C}$. Platform (15×13) was hidden 2 cm below the surface of water in a fixed location. If the rat failed to find the platform within the required time, it was guided to the platform and left there for 10 seconds. During training, animals were required to locate the submerged platform by using distal extra maze cues. Cues must be visible and useful to rats. They must be far enough away to require the rat to use spatial analysis, rather than association, to solve the task. But near enough and positioned appropriately for the rat to see and use them.



Fig. 1: The Morris Water Maze apparatus used in the present study.

Time on the platform must be sufficient for them to feel the location and to see where they are.

Two trials were given on 1st day. Cut off time for each trial was 2 minutes. 1st trial for training and 2nd trial for evaluating short term memory. Interval between trials was important. 2nd trial was performed after 60 minutes. Long-term memory was measured after 24 hours.

Neurochemical Estimations:

Tryptophan, 5HT, 5HIAA in brain and tryptophan in plasma were made by HPLC-EC as standard. A 5 II Shim-Pack ODS separation column of 4.0 mm internal diameter and 150 mm length was used. Separation was achieved by a mobile phase containing methanol (14%), octyl sodium sulfate (0.023%) and EDTA (0.0035%) in 0.1M phosphate buffer of pH 2.9 at an operating pressure of 2000-3000 psi on Shimadzu HPLC LEC 6A detector at an operating potential of 0.8 volts for biogenic amines and 1.0 volts for TRP.

Statistical Analysis:

Data are presented as mean $\pm$ SD. Neurochemical and behavioral data were analyzed by two-way ANOVA. Posthoc

comparisons were made by Newman-Keuls test. P-values <0.05 were considered significant.

RESULTS

Fig.2 shows the effect of TRP supplementation on LTM and STM of male and female rats in Morris Water Maze. Male and female untreated and treated rats took same time to reach the platform in the training session (Fig. 2a). TRP treatment did not have any effect on STM of male rats when compared with their respective controls whereas TRP treatment for 6 weeks significantly enhanced (54%) STM of female rats from their controls. Regarding the sex difference, untreated female rats exhibited better performance (31.6%) than untreated male rats. This effect on STM was more enhanced after TRP treatment. After treatment, decrease in time to reach the platform was greater (64.7%) (Fig.2b). When measuring the LTM untreated female rats exhibited a nonsignificant increase in latency time than untreated male rats. This effect was however not seen after TRP treatment. TRP treatment significantly improved LTM in male and female rats compared to controls (Fig 2c).

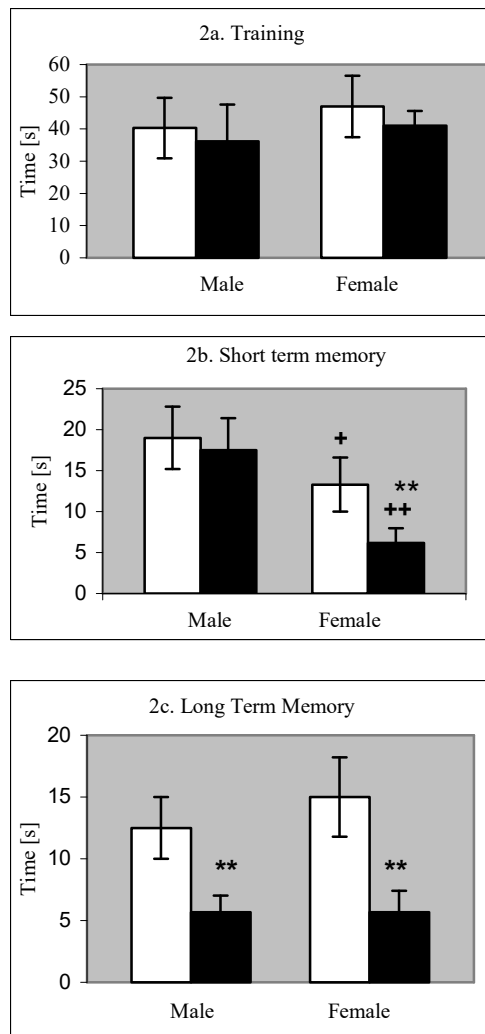


Fig. 2: Effect of TRP supplementation on Morris Water Maze test in water (white) and TRP treated (black) male and female rats. Values are means $\pm$ SD (n=6). Significant differences by Newman-Keuls test: **p<0.01 from respective water treated controls; +p<0.01, ++p<0.01 from respective male rats following two-way ANOVA.

Fig. 3 shows the effect of long-term TRP administration on the plasma TRP, brain TRP, 5HT and 5HIAA levels of male and female rats. Female rats have significantly higher levels of 5HT metabolism than in male rats.

TRP supplementation increased 5HT metabolism in both male and female rats but increases were greater in female rats than in male rats.

DISCUSSION

Administration of 100mg/kg TRP orally given to rats for 6 weeks increased plasma TRP, brain TRP, 5HT and 5HIAA levels (Markus *et al.*, 2002) and therefore, enhanced memory of rats. TRP crosses blood brain barrier predominantly by the carrier system for long chain neutral amino acids. Serotonin synthesis depends on brain TRP that in turn depends on blood concentration. Elevating TRP in the brain produces physiologically important changes in serotonergic system.

The present study examined sex differences in spatial working memory in rats. Long term and short term memory were tested in Morris water maze in which the motivating stimulus was a hidden fixed platform. This task requires subject to use the spatial arrangements of cues outside of the rectangular pool to swim to hidden fixed located platform. Male and female untreated and treated rats took same time in finding the platform during the training session (Fig. 2a). After 1 hour, when examining short term memory of rats, untreated female rats exhibited better performance (31.6%) than untreated male rats which shows that female rats had better knowledge of the precise location of platform after first exposure to the Morris water maze. Several studies have also shown a sex difference in memory functions favoring males (Seymour *et al.*, 1996; Williams *et al.*, 1990), other favoured females (Bimonte *et al.*, 2000) or no sex difference was reported (Bucci *et al.*, 1995). There are, of course, a number of possible explanations for these inconsistencies, including differences in the apparatus used and details of the task used.

After TRP treatment, the percentage decrease in time to reach the platform was greater in female (64.7%) than in male rats. TRP treatment did not have any effect on STM

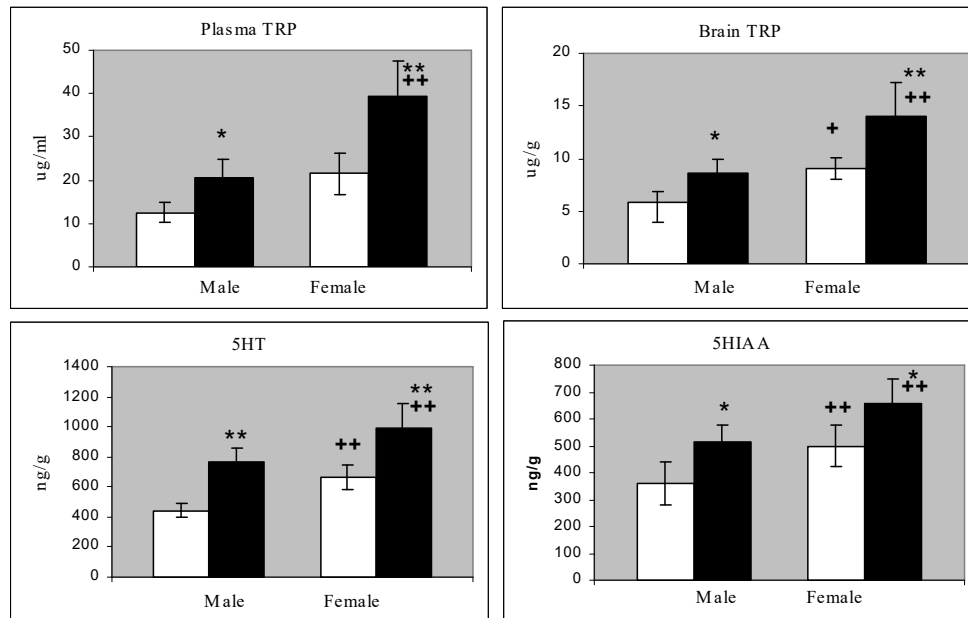


Fig. 3: Effect of TRP supplementation on plasma TRP, brain TRP, 5HT and 5HIAA levels in water (white) and TRP treated (black) male and female rats. Values are means + SD (n=6). Significant differences by Newman-Keuls test: * $p < 0.05$, ** $p < 0.01$ from respective water treated controls; + $p < 0.05$, ++ $p < 0.01$ from respective water or TRP treated male rats following two-way ANOVA.

of male rats when compared with their respective controls whereas TRP treatment for 6 weeks significantly enhanced STM of female rats compared to their corresponding controls (54%) which shows that the learning ability of female rats was more enhanced than male rats after TRP treatment (Fig 2b). A possible interpretation for the sex difference would be that males and females rely on differential strategies that are based on different representations when searching for the platform. Additionally, Sandstrom *et al.* 1998 also demonstrated that males and females differ in the kind of cues they use in tasks of spatial navigation like the Morris water maze.

After 24 hours, untreated female rats exhibited a nonsignificant increase in the latency time than untreated male rats, which shows that although females learned quicker

than male rats but they were unable to consolidate what they had learned. They spent longer time in swimming and used inefficient strategies. This tendency of impairment of memory in female rats was however not seen after TRP treatment which strengthens our previously reported suggestion that increased 5HT metabolism enhances memory functions in rats (Khaliq *et al.*, 2006). Both male and female rats exhibited comparable memory performance after TRP treatment (Fig. 2c).

Administration of TRP increased plasma and brain TRP concentration. Our study showed that administration of 100mg/kg TRP produced a rise in plasma and brain TRP. This lead to an increase in brain 5HT synthesis. Previous studies showed that a rise in brain TRP does play a role in 5HT synthesis and also determines the ongoing rate of 5HT

synthesis (Haleem, 1990; Jequir *et al.*, 1969). Several studies have reported the precursor relationship of the aminoacid L-TRP to serotonin synthesis (Curzon and Knott, 1977; Schaechter and Wurtman, 1989). As expected, in our present study also administration of TRP for 6 weeks enhanced plasma TRP, brain TRP, 5HT and 5HIAA levels in both male and female rats. Previous studies show that 5HT metabolism and turnover is greater in female (10-20%) than in male rats (Haleem *et al.*, 1990). In the present study, female rats have significantly higher levels of 5HT than in male rats. TRP supplementation increased 5HT levels in both male and female rats but increases were greater in female than in male rats.

In summary, the present results strengthen previously reported suggestions that increase 5HT metabolism enhances memory functions. In the present study we report a sex difference in the memory functions in rats. The present results also show that this effect of TRP supplementation on memory is more enhanced in females than in male rats.

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